



Oscar Moreno

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Skills

- **Frontend:** React | React Router DOM | React Hooks | React Query | Zustand & Redux | Axios | Zod | Next.js
- **Backend:** Node.js | Express | GraphQL | Jest | Supertest | Git
- **TypeScript Experience:** TypeScript (Node) | TypeScript (React)
- **Databases:** MongoDB | PostgreSQL | MySQL
- **ORMs:** Prisma | Sequelize | Mongoose
- **Programming Languages:** Python | Django | Java | C++ | C | JavaScript | D3.js
- **Computer Vision & Data Analysis:** OpenPose | OpenCV | Digital Image Processing | Tableau
- **Other Areas:** Human-Computer Interaction | Distributed Systems | Frontend | Backend | Full-Stack

Experience

Research Technician

[CTI](#)

01/2019 - Current

- Professional with experience in technical research in the academic field, with a significant focus on software development, web development (both backend and frontend), and data analysis in the field of data science. **Projects include:**

Legislatio

- Developed an interactive data visualization tool that allowed users to analyze legislative voting patterns in Ecuador, measured by participation and feedback from 13 political participants. Used **Python** to extract 2063 votes from the assembly and **D3.js, JavaScript, React** for developing the web interface.
- The web interface enabled users to intuitively explore legislative voting data and draw their own conclusions.
- Validated the benefits of the tool through testing with political actors, confirming its usefulness for understanding legislative narratives, legislative processes, and forms of citizen participation, encouraging further exploration of these objectives.

Mentorship Navigator: Visual exploration of Academic Lineages

- Developed an interactive data visualization tool that allowed users to explore academic lineages. Using **Python** to obtain 256769 authors from academic sources, I integrated the information into the tool. Implemented the user interface using **JavaScript** with **D3.js** to generate dynamic visualizations.

Bibliometric Analysis in Scientific Conferences

- Achieved bibliometric and topic analysis of the VINCI and WorldCIST conferences using Python and external libraries such as OpenAlex, AnyStyle, **Python NLTK**, and Cytoscape, as well as Mallet, to efficiently collect, process, and analyze academic data. Integrated multiple sources and optimized metadata extraction, identifying key trends and relationships in research on communication and visual interaction, as well as in the field of information systems and technologies. This study covered a set of 364 papers, providing a solid foundation for bibliometric analysis.

Ecuadorian political landscape

- Developed an interactive visualization tool in D3.js for comparative analysis of the 2021 Ecuadorian presidential election results.

Education

Bachelor Degree

Escuela Superior Politécnica del Litoral

Guayaquil, Ecuador 02/2013 - 09/2018

- Computer Science

Projects

- See my [portfolio](#)

Mentorship

- Academic assistant at ESPOL in the subjects: Web Development and Data Structures.